Final Project Proposal

1. Group Information

Members: Dawn Andrei Pamesa, Rogelio Porras, John Ashley Alba

Proposed Project Title: Monte Carlo Simulation of Threshold Bunching in Procurement Awards and How Well a Self-Baseline Test Detects It

2. Chosen Simulation Approach

[ ] Agent-Based / Grid-Based [ ] Discrete-Event Simulation [ ] Continuous / Mathematical Simulation

[X] Monte Carlo Simulation [ ] Other: _____

3. Proposed System

Our group will simulate the contract awards of one government agency over two years. Each award amount is drawn at random from a lognormal distribution. In some runs, a share of the awards that land just above the PHP 1,000,000 procurement threshold is moved to just below it, which is called bunching. We then test each simulated agency with a simple comparison of its two years: is the share of awards just below the threshold higher in year 2 than in year 1?

4. Main Question

How do the share of awards moved below the threshold and the number of awards per year affect how often the test flags the agency? How often does it raise a false alarm when nothing was moved?

5. Inputs and Outputs

Inputs or Conditions to Change	Expected Outputs / Results
Number of awards per year (100, 300 and 1,000); share of near-threshold awards moved below the threshold (0, 20 and 50 percent); width of the near-threshold band (2 percent); number of simulation runs (5,000)	Detection rate (how often the test flags the agency); false alarm rate when nothing is moved; average below-threshold share in each year; spread of the test results

6. Planned MATLAB Output

Histograms of the below-threshold share with and without bunching; a line graph of the detection rate against the share moved, one line for each agency size; a bar chart comparing the three scenarios; and a heatmap of the detection rate by awards per year and share moved.

7. Three Possible Test Scenarios

Scenario	What Will Change?
Normal / Baseline	No awards are moved (0 percent). This measures the false alarm rate.
Scenario 2	20 percent of the near-threshold awards are moved just below the threshold.
Scenario 3	50 percent of the near-threshold awards are moved just below the threshold.